

Modeling 2D Marshak Waves

Understanding the physics of x-ray radiation driven supersonic
nonlinear heat waves

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Photon Transport and Energy Exchange

Scattering-free Boltzmann equation for photons (RTE):

$$\frac{1}{c} \frac{\partial I(\hat{\Omega}, \vec{r}, t)}{\partial t} + \hat{\Omega} \cdot \vec{\nabla} I(\hat{\Omega}, \vec{r}, t) = \sigma'_a \left(B(T_m(\vec{r}, t)) - I(\hat{\Omega}, \vec{r}, t) \right) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c} \frac{\partial T_m}{\partial t} = \sigma_a(T_m) \left(\frac{1}{c} \int_{4\pi} I d\hat{\Omega} - a T_m^4 \right) \quad (2)$$

Gray Coupled Equations and LTE

We get the "gray" coupled equations:

$$\begin{aligned}\frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) &= c\sigma(U - E), & U &= aT_m^4, & E &= aT_R^4 \\ \frac{\partial e}{\partial t} &= c\sigma(E - U), & e &= fT_m^\beta \rho^{-\mu}\end{aligned}$$

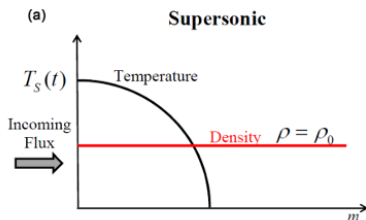
Finally, assuming complete thermodynamic equilibrium (CTE) between the radiation field and the material ($T_R = T_m$):

$$\frac{\partial T^4}{\partial t} + f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right) \quad (3)$$

Assuming $\rho f T^\beta \rho^{-\mu} \gg aT^4$ (True until around 3keV):

$$f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right)$$

Motivation



Shussman et al. POP, (2015)

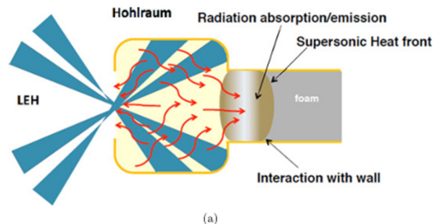
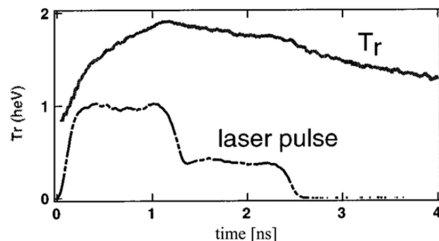
- Given the material properties equations:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

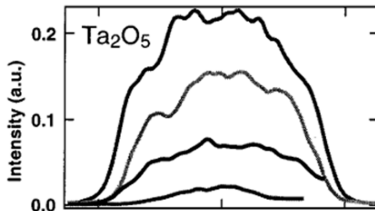
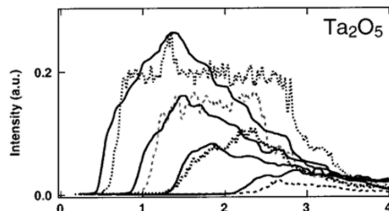
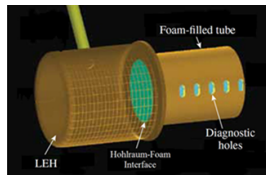
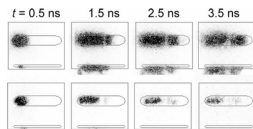
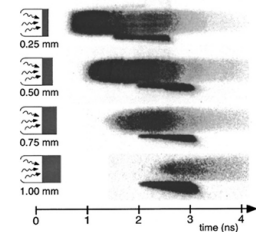
- The diffusion coefficient $D = \frac{c}{3\rho\kappa}$ is not linear.
- The entire system becomes highly non-linear and hard to solve analytically.
- Our Motivation:** To build a robust model and fundamentally understand the dynamics of this system.

Marshak Waves Experiments

Using a Hohlraum as an x-ray generator, a supersonic heatwave propagates in the low-density foam.



Marshak Waves Experiments



Experiments tested

The experiment	foam type	density (mg/cc)	temperature (eV)
Massen <i>et al.</i>	$C_{11}H_{16}Pb_{0.3852}$	80	100–150
Back <i>et al.</i>	SiO_2 , Ta_2O_5	10–50	85–190
Xu <i>et al.</i>	C_6H_{12} , $C_6H_{12}Cu_{0.394}$	50	160
Ji-Yan <i>et al.</i>	C_8H_8	160	175
Rosen and Keiter <i>et al.</i>	$C_{15}H_{20}O_6$, $C_{15}H_{20}O_6Au_{0.172}$	45–70	200–210
Moore <i>et al.</i>	SiO_2 , C_8H_7Cl	90–120	310
Courtois <i>et al.</i>	SiO_2	19 – 29	170

1.5D model

HR Model - Governing Equations

The "Black" diffusion equation (neglecting radiation energy density $\frac{\partial E}{\partial t} \rightarrow 0$):

$$\underbrace{\frac{\partial E}{\partial t}}_{\rightarrow 0} + \frac{\partial e}{\partial t} = \frac{\partial}{\partial x} \left(\frac{c}{3\rho\kappa} \frac{\partial(aT^4)}{\partial x} \right)$$

With specific energy and opacity power laws:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

where $f, \mu, g, \lambda, \alpha, \beta$ are material-dependent parameters.

HR solved the diffusion equation using perturbation theory and found some interesting results.

HR Model - Heat Front Solution

The heat front position $x_F(t)$ is given by:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

where:

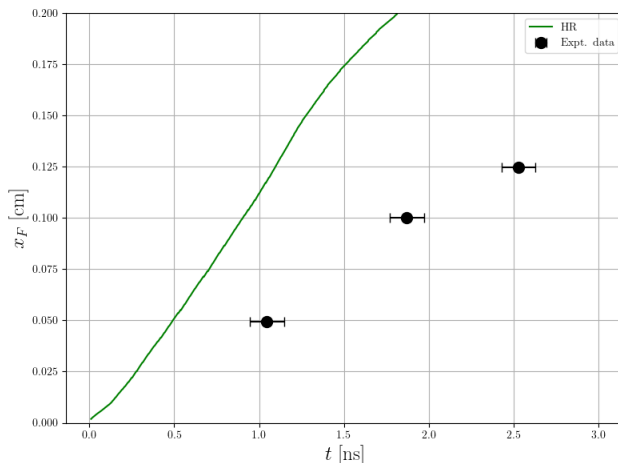
$$C = \frac{16g\sigma_{SB}}{3f(4 + \alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4 + \alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where T_s is the surface temperature at $x = 0$.

HR 1D Model Limitations

HR misses the real experimental results of Back et al. by a factor of 2!



1.5D model

1D effects

Marshak Boundary Condition: $T_D \neq T_s$

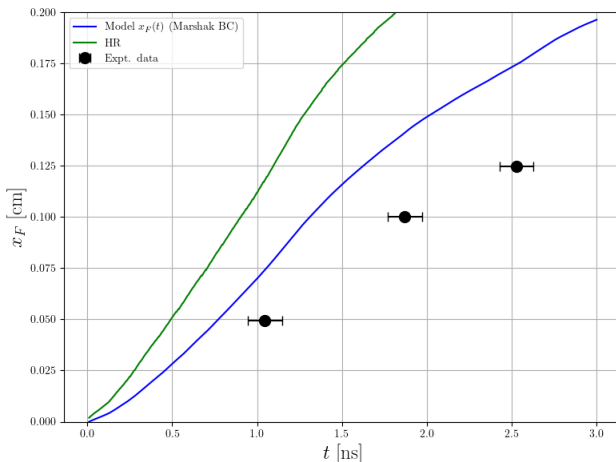
- Assuming $T_s = T_D$ overestimates the wave speed.
- Instead, energy conservation at the boundary requires:

$$\sigma_{SB} T_D^4(t) = \sigma_{SB} T_s^4(t) + \frac{F(0, t)}{2}$$

where $F(0, t)$ is the net flux entering the foam.

Marshak Boundary Condition: $T_D \neq T_s$

Applying the Marshak boundary condition slows the propagation front significantly:



1.5D model

2D effects

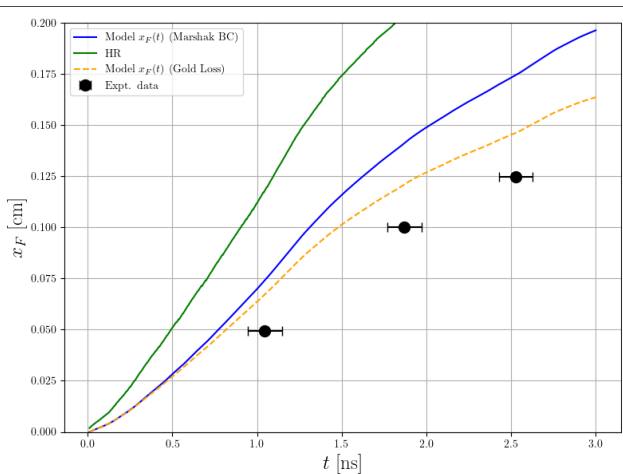
- Energy is lost from the foam into the tube walls.
- Can be modeled for different wall materials: **Gold**, **Copper**, **Beryllium**, and **Vacuum** .
- For **Gold**:

$$E_{\text{gold}}(t) = 0.59 T_s^{3.35} [t - t_0(x)]^{0.59} \quad [\text{hJ/mm}^2]$$

where $t_0(x)$ is the front arrival time at position x .

Energy Wall Loss

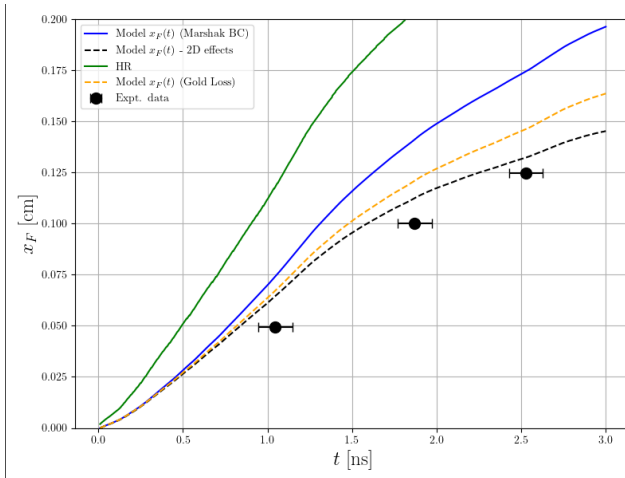
Including wall loss reduces the front velocity:



- High-Z wall material (e.g. Gold) absorbs energy and ablates inward into the foam.
- **Changing Foam Radius:** The inlet and tube radius shrink, reducing the heat wave cross-section.
- **Changing Foam Density:** The ablated wall material compresses and densifies the adjacent foam.
- Both effects must be incorporated to model the dynamic tube geometry.

Wall Ablation

Inward wall ablation compressively densifies the foam, which significantly slows down propagation:



Effective Mean Free Path

When $\lambda_R \geq 2R$, wall collisions control the transport instead of diffusion. Can be modeled as (*Courtois et al. POP, (2024)*):

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}$$

where λ_R is the Ross Mean Free Path.

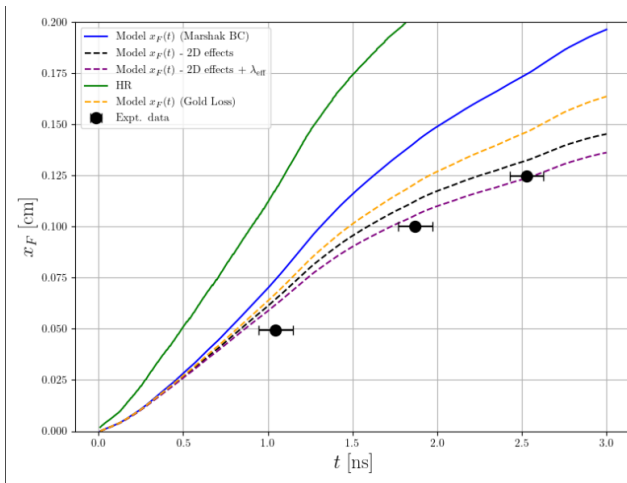
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = ((2R)^{-n} + \lambda_R^{-n})^{-1/n}$$

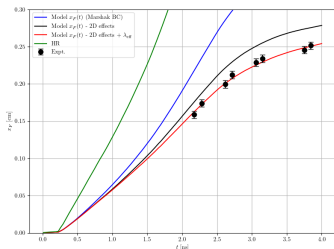
where n controls the "tightness" of the transition.

Effective Mean Free Path

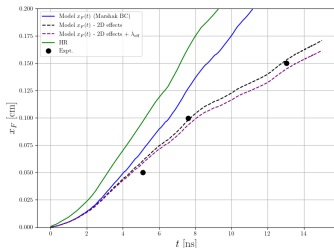
Including the effective mean free path correction matches the simulation to the experimental data at early times:



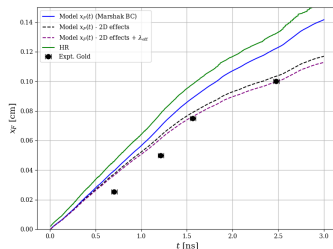
Comparison to many experiments



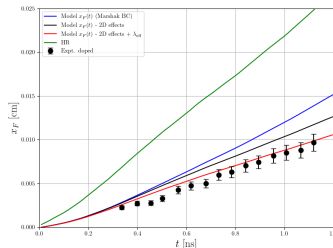
Moore et al. JQSRT, (2015)



Back et al. PRL, (2000)



Back et al. POP, (2000)



Ji-Yan et al. CPB, (2010)

Full 2D model

Modeling the front curvature

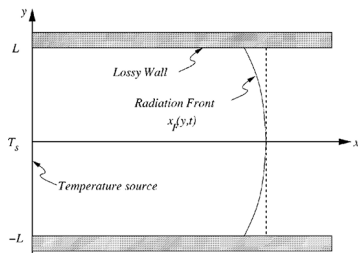
Understanding the front curvature

Hurricane & Hammer (2006) solved the laplace equation (assumed constant opacity and neglected e):

$$\underbrace{\frac{\partial e}{\partial t}}_{\rightarrow 0} = \frac{4}{3} \nabla \cdot \left(\frac{1}{\rho \kappa} \nabla (T^4) \right)$$

and got the 2D front in Cartesian coordinates (where $\varepsilon = \frac{3}{4} \rho \kappa L (1 - a)$):

$$x_F(y, t) = \frac{L}{\varepsilon} \cosh^{-1} \left(1 + \frac{D \varepsilon t}{L^2} \right) \cos \left(\frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}$$



Cylinder Coordinates

The model was re-developed in cylindrical coordinates (*Courtois et al. POP, (2022)*) to match cylindrical geometries:

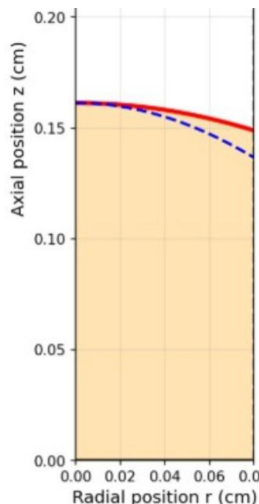
$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D\kappa_0^2 t}{2} \right) J_0(\kappa_0 r) + \text{HOT}$$

Planar $\longleftrightarrow$ Cylindrical Analogy:

$\cos(ky) \longleftrightarrow J_0(\kappa r)$ For small ε :

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4} \rho \kappa R (1 - a)$$

where a is the wall albedo.



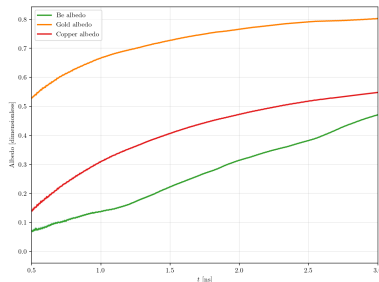
Applying on our model: Dynamic Albedo

Until now, wall albedo was treated as a constant parameter. We update it dynamically in time:

$$\alpha(t) = \frac{F_{\text{out, wall}}}{F_{\text{in, wall}}} = \frac{\sigma_{SB} T_{s, \text{interface}}^4 - \frac{dE_{\text{wall}}}{dt} / 2}{\sigma_{SB} T_{s, \text{interface}}^4 + \frac{dE_{\text{wall}}}{dt} / 2}$$

where the curvature goes as

$$J_0 \left(\frac{\sqrt{2\varepsilon(t)}}{R} r \right), \text{ and } \varepsilon(t) \text{ scales with } (1 - \alpha(t)).$$



**Problem: the longitudinal
front is way off!**

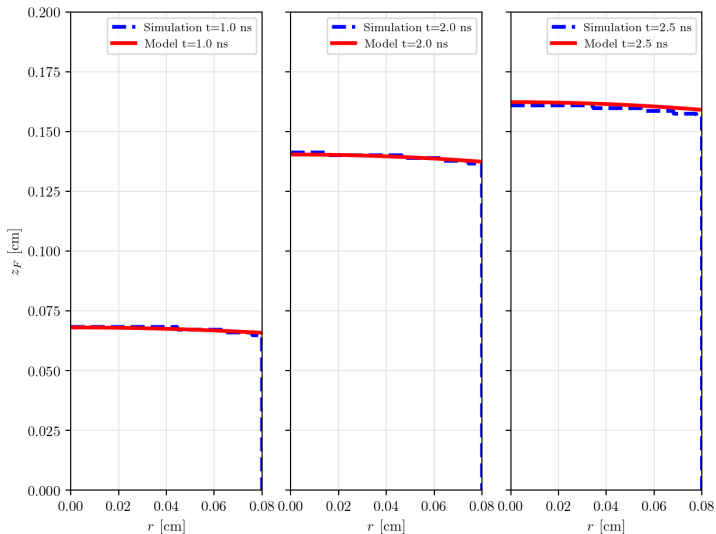
**Luckily, we have a direct
longitudinal position:**

$$x_F(r, t) = x_{1.5D}(t) \cdot J_0(\kappa_0(t)r)$$

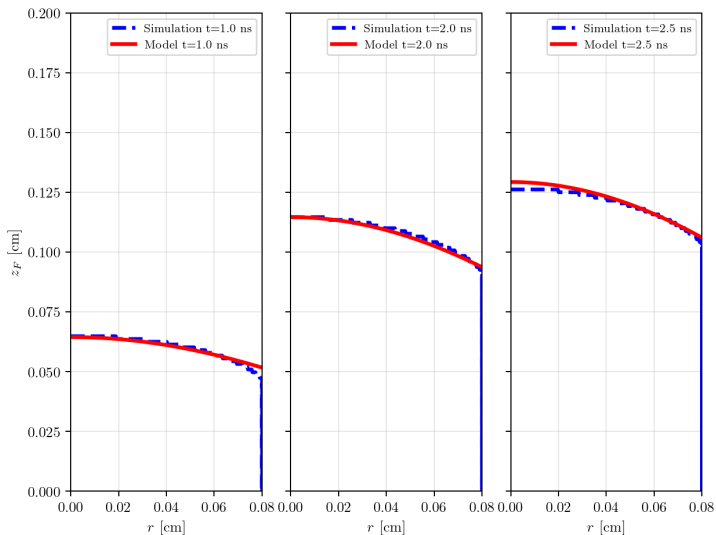
Full 2D model

Comparing to 2D supersonic Diffusion Simulation

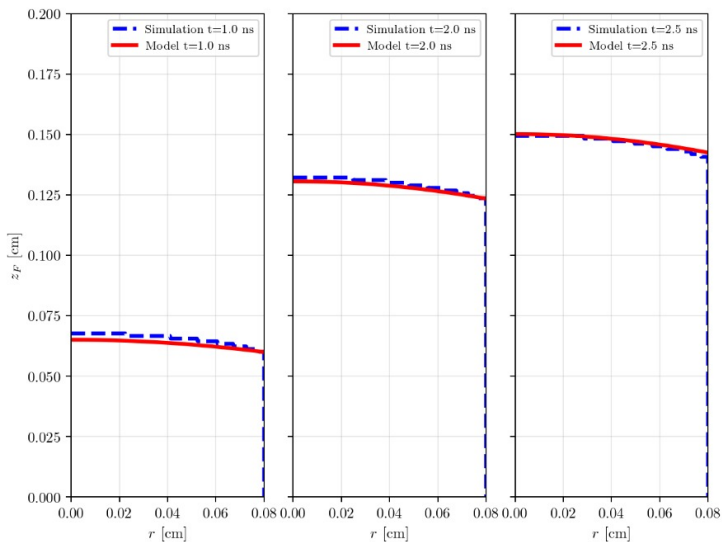
Gold coating - SiO₂

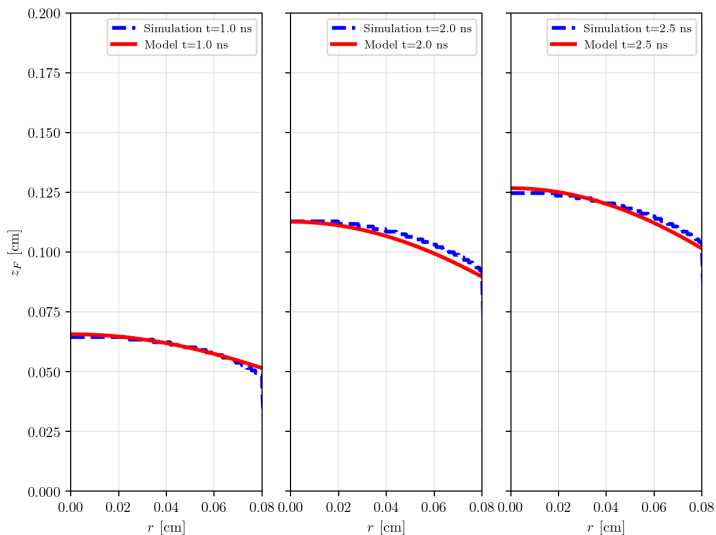


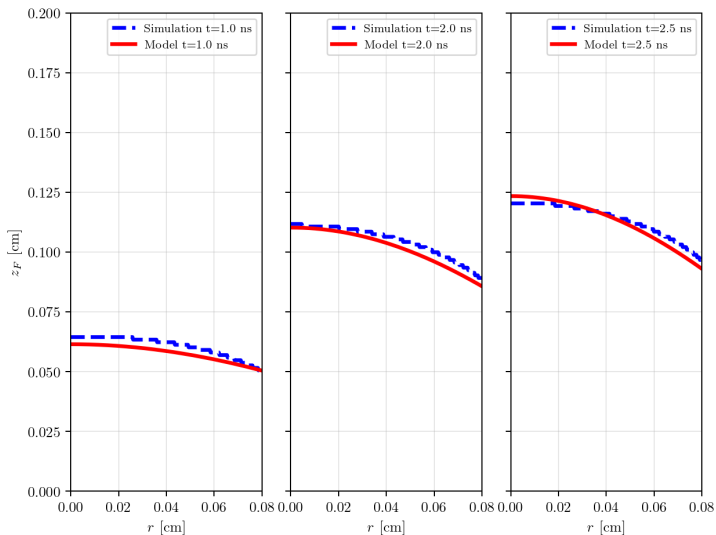
100 times transparenter Gold coating - SiO₂



Copper - SiO₂

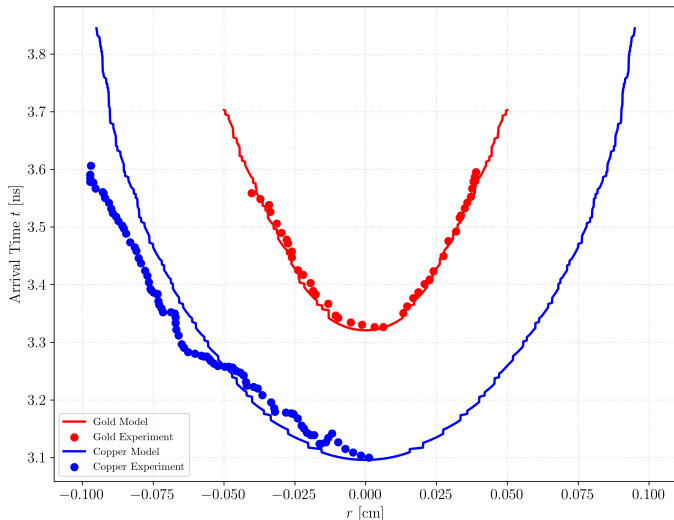






Gold vs. Copper Coating Comparison

French experiment *Courtois et al. POP, (2024)* which tests the arrival time of the front as a function of r for both gold and copper coats:



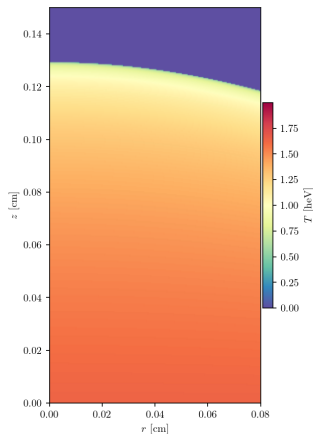
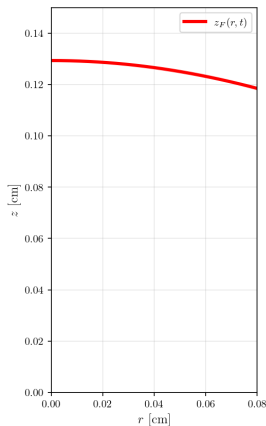
Full 2D model

A full 2D Temperature distribution

A full 2D Temperature distribution

One can use the Heyney profile:

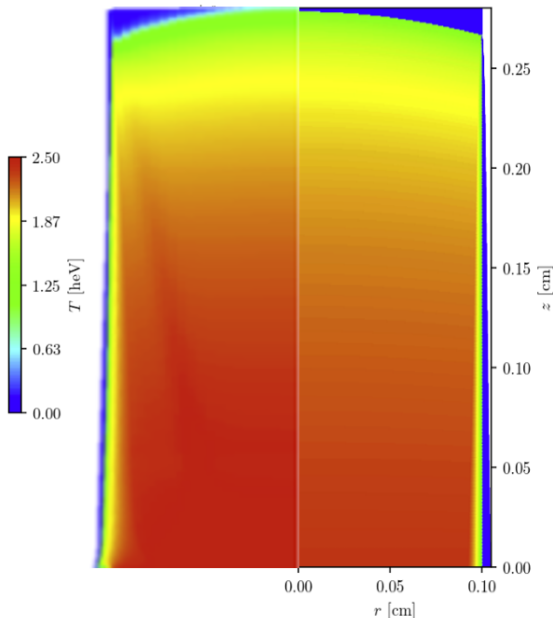
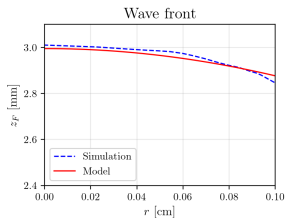
$$T(z, r, t) \approx T_S(t) \left[1 - \frac{z}{z_F(r, t)} \right]^{\frac{1}{4+\alpha-\beta}}$$



2D Temperature Distribution — Moore (C_8H_7Cl)

Side-by-side comparison of the 2D temperature distribution at $t = 3.72$ ns:

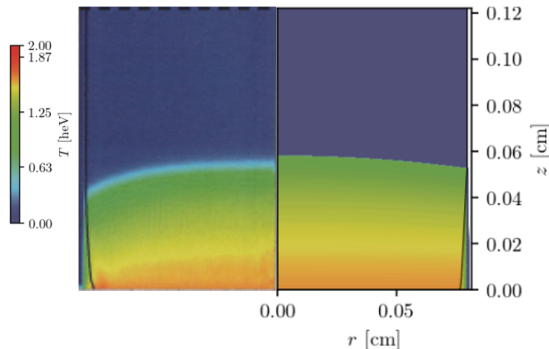
- **Left:** Diffusion simulation taken from *Moore et al. JQSRT, (2015)*.
- **Right:** Our full 2D model.



2D Temp. Distribution — Back SiO₂ (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

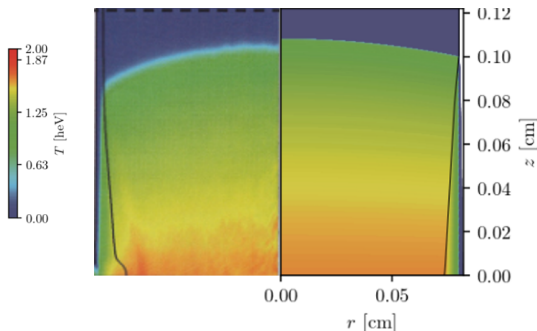
- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Back SiO₂ (2.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.0$ ns:

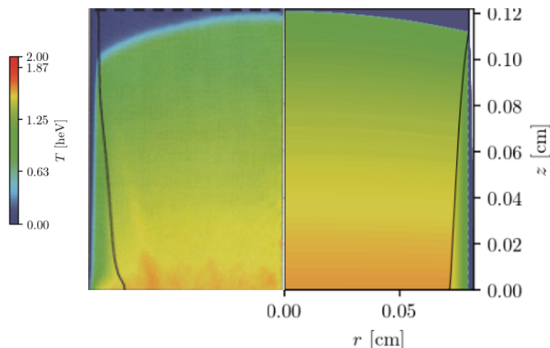
- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
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2D Temp. Distribution — Back SiO₂ (2.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.5$ ns:

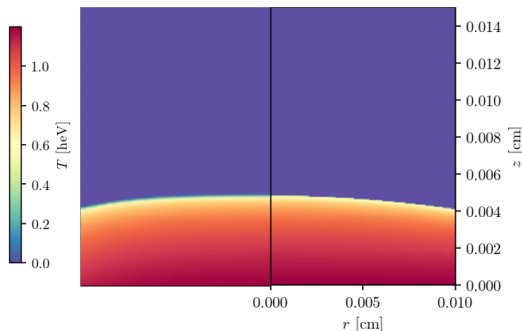
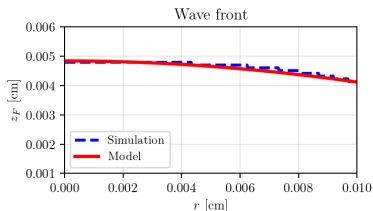
- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Ji-Yan (0.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 0.5$ ns:

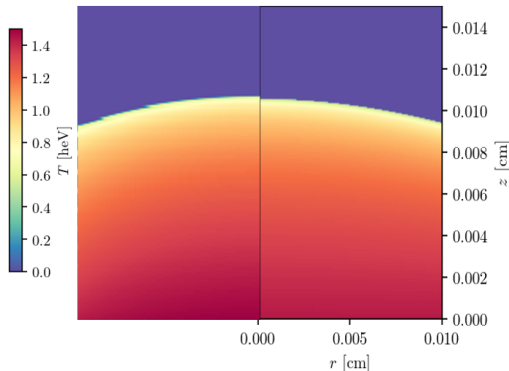
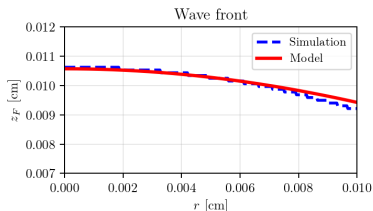
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D Temp. Distribution — Ji-Yan (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

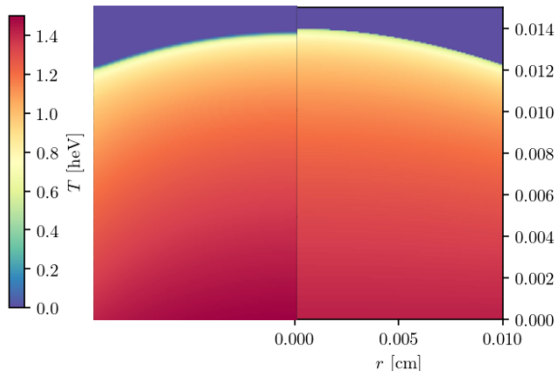
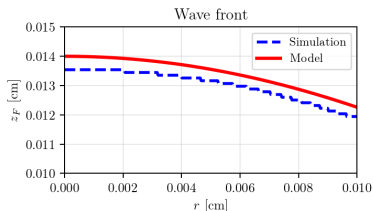
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



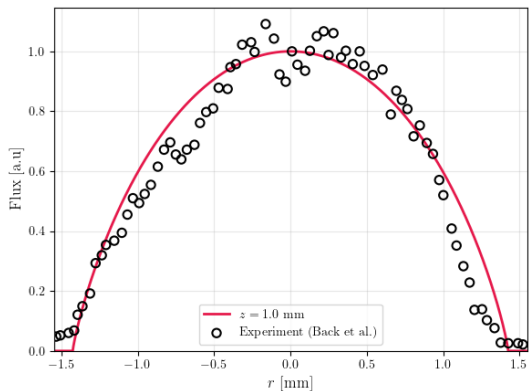
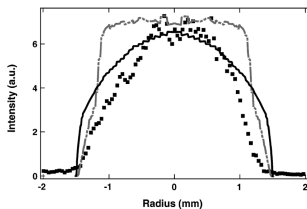
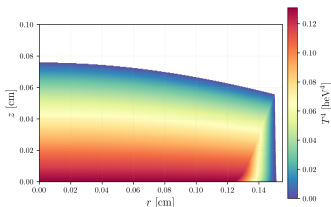
2D Temp. Distribution — Ji-Yan (1.3 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.3$ ns:

- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.

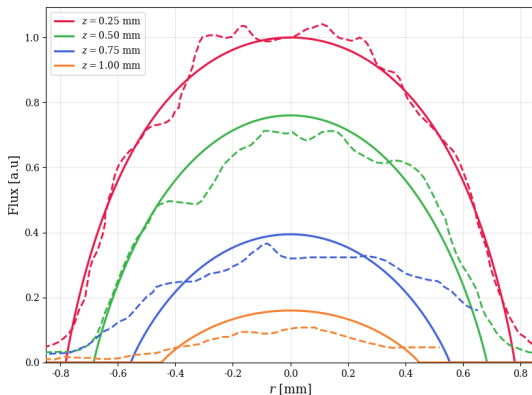
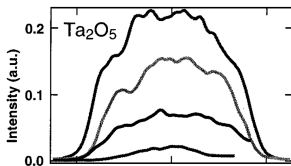
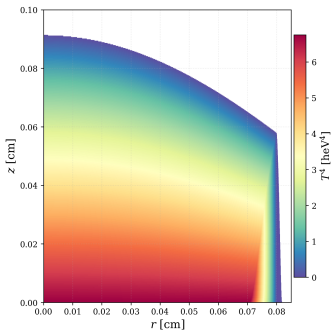


Flux Curvature Comparison - Back SiO₂ PRL



Back et al. PRL, (2000)

Flux Curvature Comparison - Back Ta2O5 POP



Back at el. POP, (2000)

- **Successful 2D Modeling:**

- We have successfully modeled 2D supersonic Marshak waves.
- Our unified model captures front curvature and temperature distribution.

- **Deep Physical Insight:**

- We can now understand the underlying physics of radiation transport and energy exchange in a very good, intuitive way.

Thank You!

Questions?